

Docker + Nginx Deployment on AWS EC2 - Step by Step

1. SSH: `ssh -i devops-key.pem ubuntu@YOUR_PUBLIC_IP`

2. Update:

```
sudo apt update
sudo apt upgrade -y
```

3. Docker:

```
sudo apt install docker.io -y
sudo systemctl enable docker
sudo systemctl start docker
sudo systemctl status docker
```

4. Verify:

```
docker --version
```

5. Permissions:

```
sudo usermod -aG docker ubuntu
newgrp docker
docker ps
```

6. Project:

```
mkdir docker-web
cd docker-web
```

7. Copied the index.html file from local machine using (`scp -i key source destination`)

8. Dockerfile:

```
FROM nginx:latest
COPY index.html /usr/share/nginx/html/index.html
EXPOSE 80
```

9. Build: `docker build -t myweb:v1 .` docker images

10. Run:

```
docker run -d --name web-container -p 8000:80 myweb:v1
```

11. Verify:

```
docker ps
```

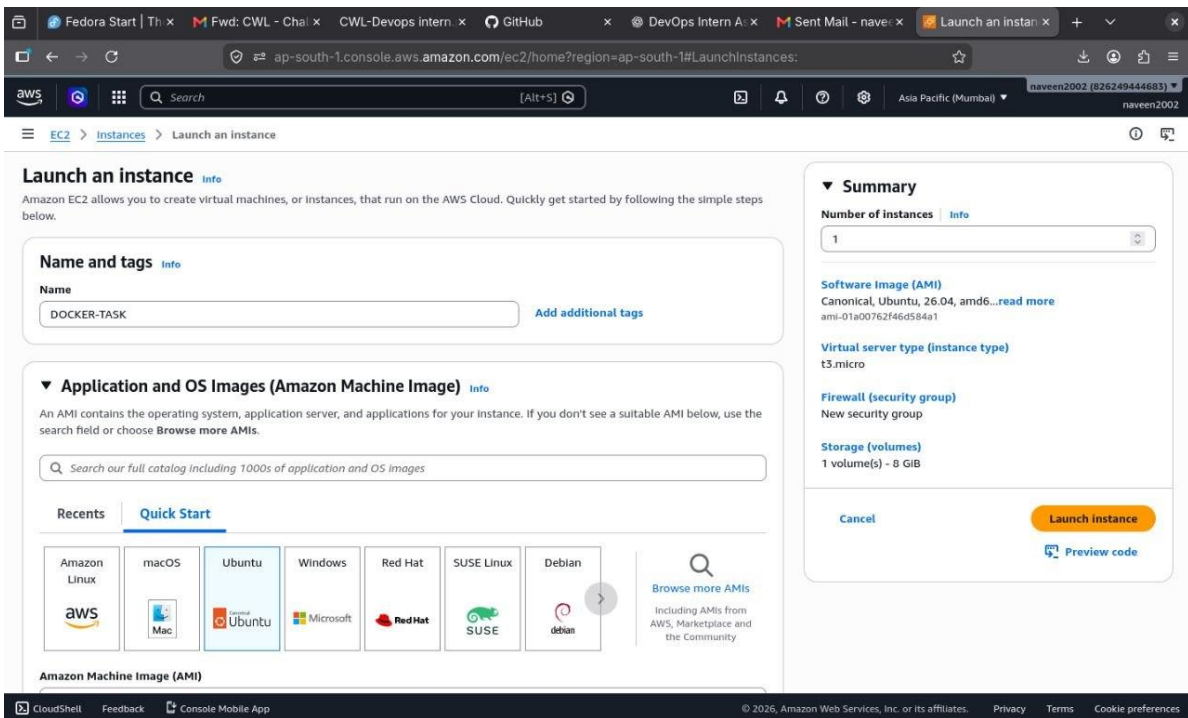
12. Security Group: Custom TCP 8000 from 0.0.0.0/0

13. Browser: `http://YOUR_PUBLIC_IP:8000`

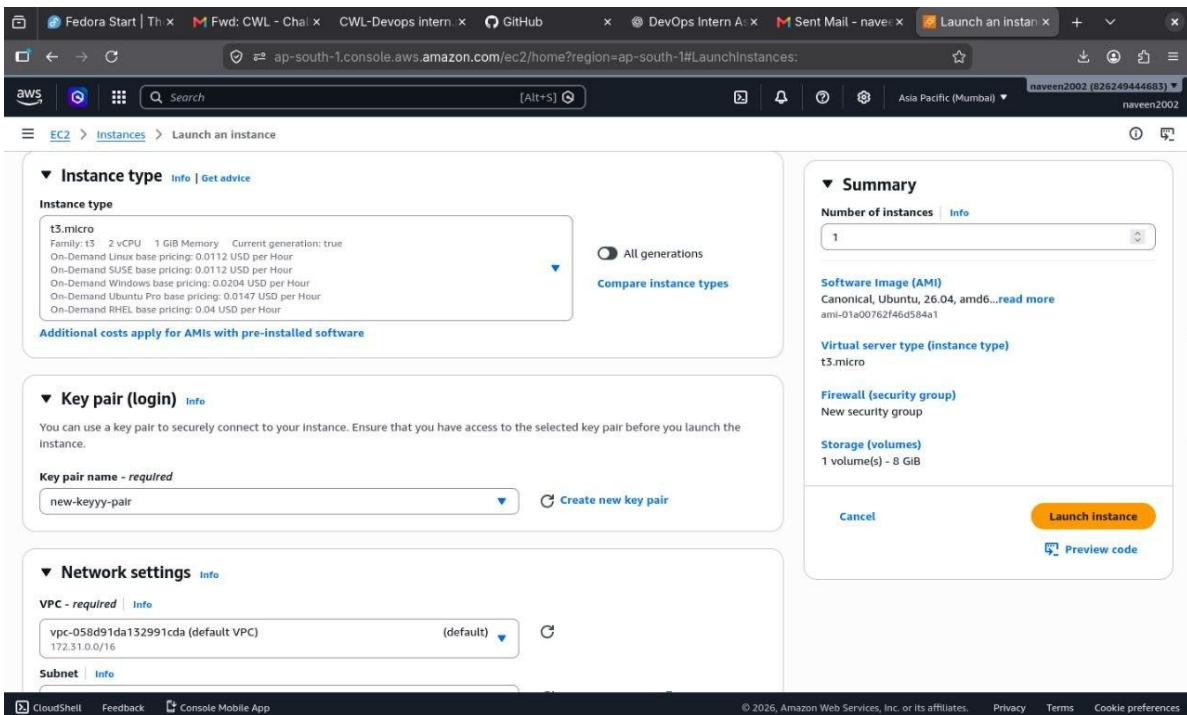
Useful	Commands:	
images	<code>docker ps</code>	<code>docker</code>
stop web-container	<code>docker</code>	
start web-container	<code>docker</code>	
logs web-container	<code>docker</code>	
rm web-container	<code>docker</code>	
rmi myweb:v1		

ADD THE ALL SCREENSHOTS ALSO

1.



2.



3.

The screenshot shows the 'Launch an Instance' wizard in the AWS Management Console, specifically the 'Network settings' step. The page is for the 'ap-south-1' region. The 'VPC' is set to 'vpc-058d91da132991cda (default VPC)'. The 'Subnet' is set to 'No preference'. The 'Availability Zone' is set to 'No preference'. The 'Auto-assign public IP' is set to 'Enable'. The 'Firewall (security groups)' section shows 'Create security group' selected, with a 'Security group name' of 'launch-wizard-1' and a 'Description' of 'launch-wizard-1 created 2026-07-11T03:53:44.300Z'. The 'Summary' panel on the right shows 'Number of instances' as 1, 'Software image (AMI)' as 'Canonical, Ubuntu, 26.04, amd64', 'Virtual server type (instance type)' as 't3.micro', 'Firewall (security group)' as 'New security group', and 'Storage (volumes)' as '1 volume(s) - 8 GiB'. The 'Launch instance' button is highlighted in orange.

Network settings

VPC - required

vpc-058d91da132991cda (default VPC) (default)

Subnet

No preference

Availability Zone

No preference

Auto-assign public IP

Enable

Firewall (security groups)

Create security group

Select existing security group

Security group name - required

launch-wizard-1

Description - required

launch-wizard-1 created 2026-07-11T03:53:44.300Z

Inbound Security Group Rules

Summary

Number of instances

1

Software image (AMI)

Canonical, Ubuntu, 26.04, amd64...read more

ami-01a00762f46d584a1

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

4.

The screenshot shows the 'Launch an Instance' wizard in the AWS Management Console, specifically the 'Instance type' step. The 'Instance type' is set to 't3.micro'. The 'Key pair (login)' section shows 'new-keyyy-pair' selected. The 'Network settings' section is partially visible at the bottom. The 'Summary' panel on the right shows 'Number of instances' as 1, 'Software image (AMI)' as 'Canonical, Ubuntu, 26.04, amd64', 'Virtual server type (instance type)' as 't3.micro', 'Firewall (security group)' as 'New security group', and 'Storage (volumes)' as '1 volume(s) - 8 GiB'. The 'Launch instance' button is highlighted in orange.

Instance type

t3.micro

Family: t3 2 vCPU 1 GiB Memory Current generation: true

On-Demand Linux base pricing: 0.0112 USD per Hour

On-Demand SUSE base pricing: 0.0112 USD per Hour

On-Demand Windows base pricing: 0.0204 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0147 USD per Hour

On-Demand RHEL base pricing: 0.04 USD per Hour

Additional costs apply for AMIs with pre-installed software

Key pair (login)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

new-keyyy-pair

Network settings

VPC - required

vpc-058d91da132991cda (default VPC) (default)

Subnet

No preference

Summary

Number of instances

1

Software image (AMI)

Canonical, Ubuntu, 26.04, amd64...read more

ami-01a00762f46d584a1

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

5.

The screenshot shows the 'Launch an instance' wizard in the AWS Management Console, specifically the 'Network settings' page. The page is for the 'ap-south-1' region. The 'VPC' is set to 'vpc-058d91da132991cda (default VPC)'. The 'Subnet' is set to 'No preference'. The 'Availability Zone' is set to 'No preference'. The 'Auto-assign public IP' is set to 'Enable'. The 'Firewall (security groups)' section shows 'Create new security group' selected, with a name 'launch-wizard-1' and a description 'launch-wizard-1 created 2026-07-11T05:53:44.300Z'. The 'Summary' panel on the right shows 'Number of instances' as 1, 'Software Image (AMI)' as 'Canonical, Ubuntu, 26.04, amd64...', 'Virtual server type (instance type)' as 't3.micro', 'Firewall (security group)' as 'New security group', and 'Storage (volumes)' as '1 volume(s) - 8 GiB'. At the bottom, there are 'Cancel' and 'Launch instance' buttons, along with a 'Preview code' link.

6.

```
ubuntu@ip-172-31-34-169: ~ -- sudo su
file:///2002/home/naveen/Downloads

individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ubuntu@ip-172-31-34-169:~$ sudo apt upadte
Errors: Invalid operation upadte
ubuntu@ip-172-31-34-169:~$ sudo apt update -y
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease [136 kB]
Get:4 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 Packages [16.3 MB]
Get:5 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/universe Translation-en [6329 kB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:7 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64 Components [4556 kB]
Get:8 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64 c-n-f Metadata [313 kB]
Get:9 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/multiverse amd64v3 Packages [291 kB]
Get:10 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/multiverse Translation-en [127 kB]
Get:11 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/multiverse amd64 Components [50.0 kB]
Get:12 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/multiverse amd64 c-n-f Metadata [8276 B]
Get:13 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 Packages [338 kB]
Get:14 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [86.7 kB]
Get:15 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [53.1 kB]
Get:16 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64 c-n-f Metadata [5756 B]
Get:17 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64v3 Packages [202 kB]
Get:18 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [62.5 kB]
Get:19 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [153 kB]
Get:20 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 c-n-f Metadata [4704 B]
Get:21 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/restricted amd64v3 Packages [248 kB]
Get:22 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [45.5 kB]
Get:23 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64v3 Packages [3360 B]
Get:24 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/multiverse Translation-en [772 B]
Get:25 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Components [216 B]
Get:26 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 c-n-f Metadata [256 B]
Get:27 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports/main amd64 Components [212 B]
Get:28 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports/main amd64 c-n-f Metadata [112 B]
Get:29 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [216 B]
Get:30 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports/universe amd64 c-n-f Metadata [116 B]
Get:31 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports/restricted amd64 Components [216 B]
Get:32 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports/restricted amd64 c-n-f Metadata [120 B]
Get:33 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports/multiverse amd64 Components [216 B]
```

7.

```
ubuntu@ip-172-31-34-169: ~ -- sudo su
naveen@2002:~$ cd Downloads/
naveen@2002:~/Downloads$ sudo su
[sudo] password for naveen:
root@2002:/home/naveen/Downloads# ssh -i new-keyyy-pair.pem ubuntu@13.233.108.161
The authenticity of host '13.233.108.161 (13.233.108.161)' can't be established.
ED25519 key fingerprint is SHA256:xPonGW07HsqBsPet1zeVAziCNs+Fi2mwGymcCrC6/o.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '13.233.108.161' (ED25519) to the list of known hosts.
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Jul 11 03:58:46 UTC 2026

System load:  0.1          Temperature:   -273.1 C
Usage of /:   30.5% of 6.61GB  Processes:    121
Memory usage: 26%          Users logged in: 0
Swap usage:  0%            IPv4 address for ens5: 172.31.34.169

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
```

8.

```
</html>root@2002:/home/naveen/Downscp -i new-keyyy-pair.pem index.html ubuntu@13.233.108.161:/home/ubuntu/docker-project
index.html 100% 1246 15.5KB/s 00:00
root@2002:/home/naveen/Downloads#
```

```
Free Cloud Computing Services | Instances | EC2 | ap-south-1 | EC2 Instance Connect | ap-south-1 | + | Ask Gemini | - |
ap-south-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=ap-south-1&connType=standard&instanceId=i-0cabb901a3a923ee0&osUser...
aws | Search | [Alt+S] | Asia Pacific (Mumbai) | naveen2002 (82624944683) | naveen2002

*** System restart required ***
Last login: Sat Jul 11 07:20:03 2026 from 13.233.177.3
ubuntu@ip-172-31-34-169:~$ ls
docker-project
ubuntu@ip-172-31-34-169:~$ cd docker-project/
ubuntu@ip-172-31-34-169:~/docker-project$ history
1  ls
2  cd docker-project/
3  ls
4  vi Dockerfile
5  docker build -t myweb:v1 .
6  sudo su
7  sudo apt update
8  sudo apt update -y
9  sudo apt upgrade -y
10 sudo apt install docker.io -y
11 sudo systemctl enable docker
12 sudo systemctl start docker
13 docker --version
14 mkdir docker-project
15 ls
16 cd docker-project/
17 pwd
18 scp -i new-keyyy-pair.pem index.html ubuntu@13.233.108.161:/home/ubuntu/docker-project
19 ls
20 cat index.html
21 ls
22 cd docker-project/
23 history
ubuntu@ip-172-31-34-169:~/docker-project$
```

